

MODELING WELL-BEING USING ZIP CODE SPATIAL NEIGHBORHOODS UNDER A BAYESIAN BETA REGRESSION FRAMEWORK

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Introduction

Community well-being indices

- Well-being indices (WBI) are useful tools that measure health and wellness across various dimensions
- Examples:
 - AARP's *Livability Index*
 - CDC's *Social Vulnerability Index*
 - Robert Wood Johnson Foundation's *County Health Rankings*

Modeling WBI

- Individual-level well-being data permits analysis of community health and wellness
- Measures typically computed for large geographical areas such as states and counties
 - Incorporating spatial information is done by assuming neighboring units are similar in health factors and outcomes¹
 - However, there is usually large variation in well-being within states/counties
- ZIP Codes are much smaller units and the assumption of spatial similarity is more reasonable

¹P. L. Remington, *Health Services Research* **50**, 1407 (2015).

Research goal

- Obtain more precise estimates and rankings of communities at the Zip Code-level by employing a model which allows us to “borrow” information:
 - from neighboring ZIP Codes for ZIP Codes with missing data
 - from prior years for future years

Methods

WBI and where one lives

- WBI has bounded support from 0 to 100 so we employ a GLM, specifically a Beta regression to model it

$$g(\mu_{si}) = x_{si}^T \beta + \alpha_s$$

- Higher values of α_s indicate improvement in the WBI y_{si} for the participant i due to belonging to ZIP Code s
 - Accounts for all observable and unobservable qualities of a ZIP Code that influence WBI
- ZIP Codes can be ranked based on α_s

Beta regression

We follow the Ferrari and Cribari-Neto (2004) parameterization of the Beta distribution, resulting in the following density function:

$$f(y; \mu, \phi) = \frac{\Gamma(\phi)}{\Gamma(\mu\phi)\Gamma((1-\mu)\phi)} y^{\mu\phi-1} (1-y)^{(1-\mu)\phi-1},$$

where $y \in (0, 1)$, $0 < \mu < 1$ and $\phi > 0$.

- This parameterization has the convenient property of setting $E[y] = \mu$ and $Var(y) = \frac{\mu(1-\mu)}{1+\phi}$
- ϕ serves as a precision parameter since $Var(y)$ increases as ϕ decreases, assuming that μ is held constant

Bayesian beta regression framework - mean model

$$\text{logit}(\mu_{ist}) = \frac{\mu_{ist}}{1 - \mu_{ist}} = \eta_{ist} = \mathbf{x}_{ist}\boldsymbol{\beta}_t + \alpha_{st},$$

- $\mathbf{x}_{ist}$ = $p \times 1$ vector of individual-level characteristics for participant i in zip-code s at year t
- $\boldsymbol{\beta}_t$ = $p \times 1$ vector of corresponding regression coefficients for year t
- α_{st} = ZIP Code-level spatial effect for ZIP Code s at year t

Priors:

$$\boldsymbol{\beta}_t \sim N_p(\mathbf{0}, \tau_\mu I_p)$$

$$\boldsymbol{\alpha}_t \sim N_S(\mathbf{0}, M^{-1})$$

Bayesian beta regression framework - dispersion model

$$\log(\phi_{st}) = \zeta_{st} = \omega_0 + \mathbf{c}_{st}^T \boldsymbol{\omega}_t,$$

- ω_0 = intercept
- $\mathbf{c}_{st}$ = $q \times 1$ vector of ZIP Code-level characteristics for ZIP Code s at year t
- $\boldsymbol{\omega}_t$ = $q \times 1$ vector of corresponding regression coefficients for year t

Priors:

$$\boldsymbol{\omega}_t \sim N_q(\mathbf{0}, \tau_\phi I_q)$$

Spatial neighborhood of ZIP Codes

- ZIP Codes are neighbors if the driving time between their population centroids is < 30 mins
- **Advantage:** Model borrows information from neighboring ZIP Codes
 - Useful to estimate effects for ZIP Codes with no responses
- We use a multivariate normal prior for α where the covariance is the inverse of $M = \lambda L + \lambda \gamma I_s$
 - $L = D - A$ is the graph Laplacian matrix
 - Assign minimally informative Gamma priors for λ and γ

Bayesian Beta model - subsequent years

- We use posterior estimates of α from year $t - 1$ as prior means for α for year $t \in \{2, \dots, T\}$

$$\alpha_t \sim N_S(\hat{\alpha}_{t-1}, M^{-1})$$

- ZIP Code effect estimates changes are likely minimal from year to year so setting an informative prior based on previous year's data should aid estimation

Model implementation

- Not feasible to derive posterior distribution for parameters from standard distributions
- Use Stan to run the Bayesian sampler
- Samples can be used for inference by computing posterior means/medians/modes and credible intervals

Marginal effects

- Coefficients are difficult to interpret because of logit link function
- We instead use marginal effects to evaluate effects of variables on WBI
- Assume that there are $X_1, \dots, X_p$ variables.
 - If X_1 is a continuous variable, the marginal effect of X_1 on Y is:

$$\delta = E[Y|X_1 = x_1 + 1, X_2 = x_2, \dots, X_p = x_p, \alpha] - E[Y|X_1 = x_1, X_2 = x_2, \dots, X_p = x_p, \alpha]$$

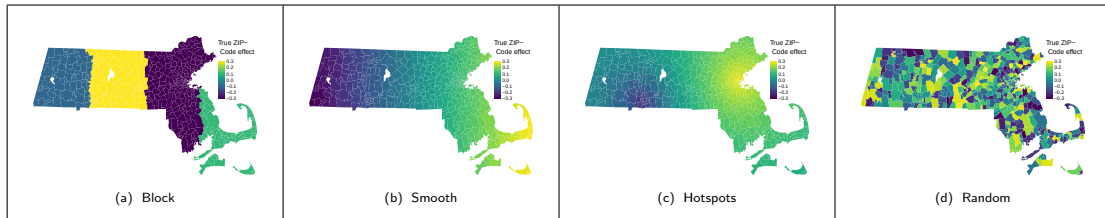
- If X_1 is a binary variable, the marginal effect of X_1 on Y is:

$$\delta = E[Y|X_1 = 1, X_2 = x_2, \dots, X_p = x_p, \alpha] - E[Y|X_1 = 0, X_2 = x_2, \dots, X_p = x_p, \alpha]$$

Simulation

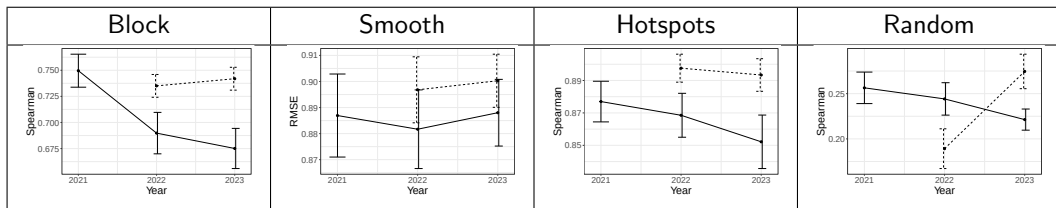
Simulation setup

- 3 individual-level variables – Sex, Age, Income (4 categories)
- 3 ZIP Code-level attributes – % College degree, % White, % Age 65+
- 3 Precision settings ($\omega_0 = 1, 3, 5$) - intercept for the Dispersion model
- 4 spatial patterns – block, smooth, hotspots, random
- 30 replications for each of the 12 simulation setting



Simulation results

- Accurate estimation of regression coefficients in both mean and dispersion models
- General spatial distribution of ZIP Code effects is captured for each spatial pattern
- Estimation of ZIP Code effects is improved in low precision settings when using informative prior for α_s and no worse in medium or high precision settings
- Plots show average Spearman correlation for all spatial patterns in the low precision setting



Case study: Massachusetts

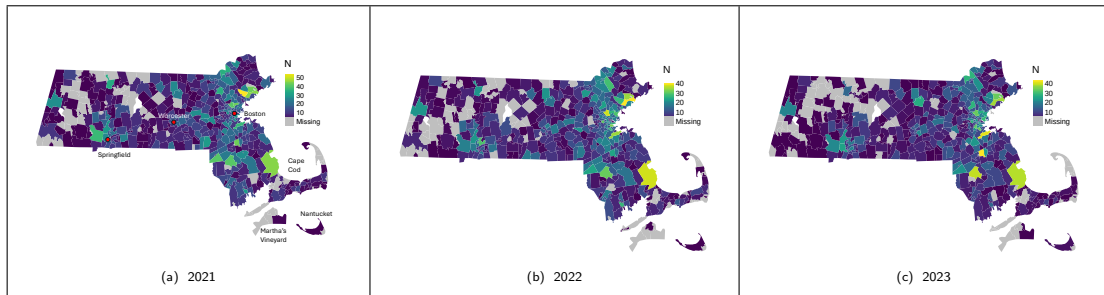
Massachusetts data

Year	Sample size	ZCTAs represented (of 537)
2021	4,443	448
2022	3,602	439
2023	3,428	435

- We used 6 individual-level variables for the mean model: Sex, Age, Race (5 categories), Marital status (3 categories), Education (4 categories), and Income (4 categories)
- Used 5 ZIP Code-level covariates for the precision model: Percent female, Age diversity, Racial diversity index, Economic residential segregation, Educational residential segregation

ZIP Code coverage

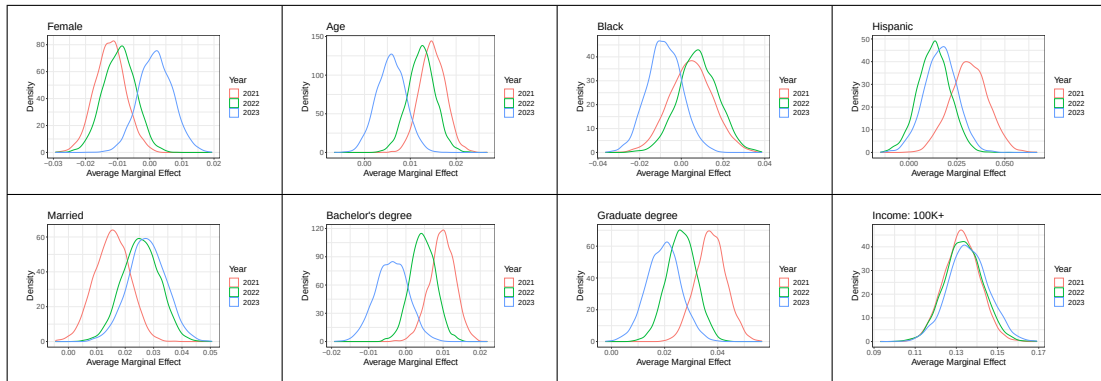
Number of observations in each ZIP Code by year. Grey ZIP Codes are ones with no observations.



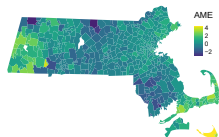
Model convergence

- 3 chains with 10,000 iterations in each chain. Burn-in of 4,000 and thinning of 5
- $\hat{R} \leq 1.01$ for all model parameters (variables, ZIP code effects, tuning parameters)

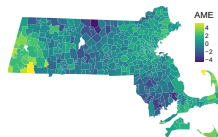
Average marginal effects



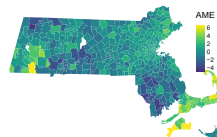
ZIP Code choropleths



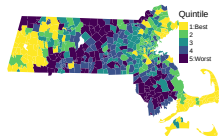
(l) 2021 ZCTA Average Marginal Effects



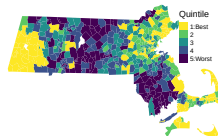
(m) 2022 ZCTA Average Marginal Effects



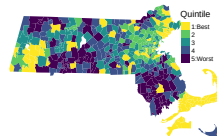
(n) 2023 ZCTA Average Marginal Effects



(o) 2021 ZCTA Quintile Rankings

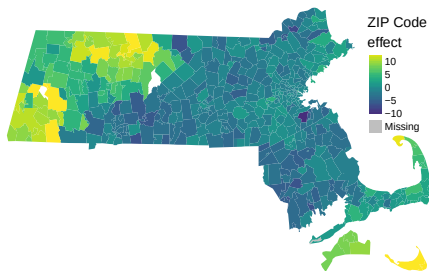


(p) 2022 ZCTA Quintile Rankings

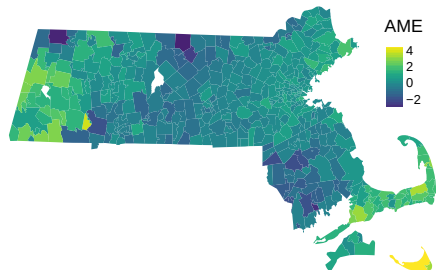


(q) 2023 ZCTA Quintile Rankings

Comparison to ZIP Code effects from previous work



(a) ZIP Code Effects from Linear spatial model (outliers trimmed)



(b) Marginal Effect of ZIP Codes from Bayesian beta regression model

Discussion

Conclusions

- We show through simulations that Bayesian double generalized beta regression framework can accurately model and capture effect sizes in both the mean and precision models under various spatial patterns and noise settings
- ZIP Code effect estimates are more stable under Bayesian Beta regression framework; in our previous work we found some outlier estimates

Limitations and future work

Limitations:

- Long computational time - not scalable for larger states or the whole country
 - Real data model for single year at 10,000 iterations with 4000 burn in takes about 2 days

Future work:

- Speeding up computational time using approximation techniques for estimation (ex. INLA)

Thank you!

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